Effects of L2 proficiency and response position on the degree of commonality in Chinese EFL learners' word associations

Abstract: Previous word association (WA) research shows that second language (L2) learners exhibit a higher degree of associative commonality – that is, produce more common WA responses – as they progress to higher proficiency levels, which is indicative of development towards a highly shared L2 lexicon. However, such conclusion can only provide a partial account for the developing L2 lexicon because the potential effect of response position has not been considered. This factor has been shown by research on first language (L1) speakers to influence the degree of associative commonality in that later (second and third) responses appear more diverse than first responses. The present study is an attempt to investigate whether such characteristic exists in L2 WAs as well. Specifically, we studied the effects of L2 proficiency and response position (1st, 2nd, and 3rd response) on the degree of commonality in WA responses of Chinese learners of English as a foreign language (EFL). A total of 50 Chinese university students – 29 from the first year and 21 from the third year – participated in the study. They were instructed to complete a web-based WA test, which contained 30 high-frequency English words. Degree of commonality was calculated on each response based on its raw occurrence in the response set to the corresponding stimulus word and this procedure resulted in a total score for each participant. A 2×2 between-subject ANOVA was conducted with degree of commonality as the dependent variable and L2 proficiency (first-year/third-year) and response position (first/second/third) as the independent variables. Results indicate a significant main effect for L2 proficiency, with third-year students producing more common responses. In addition, there is a significant main effect for response position, with later responses being more heterogeneous than earlier responses. Furthermore, a significant interaction effect of L2 proficiency and response position is also found, with associative commonality declining faster with response position in the first-year group. Overall, our findings substantiate the combined effect of language proficiency and generation position of word association responses, and we suggest future research take these factors into account in empirical investigation of the associative characteristics in the L2 mental lexicon.

INTRODUCTION

Word association (WA) functions as an easy window into the organization of words in the mind and has been widely used by second language (L2) researchers to investigate the structure of

lexical knowledge in the mental lexicon of L2 learners. From a developmental perspective, considerable efforts have been devoted to establishing the potential relationships between L2 proficiency growth and the characteristic patterns derived from the WA responses of L2 learners. Among them, two patterns dominated the past research: the native-like pattern determined by comparing L2 learners' WA responses to the norms of native speakers and the lexical-semantic pattern obtained by categorizing WA responses according to lexical relationships. Although both lines of research have yielded important findings, they have several limitations. The native-like pattern, for example, is not always reliably correlated with L2 proficiency and is affected by how native-likeness is predetermined as well as several issues associated with native WA norms. Lexical-semantic pattern, on the other hand, is directly affected by the choice of classification frameworks and widely criticized for raters' subjectivity involved in the categorization procedure. Therefore, these two patterns are far from sufficient in leading us to a complete understanding of L2 WA development, and the exploration of more reliable patterns is needed in this regard. Motivated by early psycholinguistic work and some recent endeavors to reveal different facets of L2 lexical organization, this paper attempts to investigate two important characteristic patterns of L2 WA responses: commonality and diversity. While early WA research differed greatly in the actual operationalization of commonality, it is generally defined as the degree a group of subjects generate common associative responses to a set of prompt words. Previous research demonstrates that L2 proficiency is positively correlated with associative commonality, indicating that more proficient learners seem to have more shared concepts in their L2 lexicon. Closely related is the notion of diversity, which is defined as the number of different associative responses by a group of subjects. To date, little research has been done to systematically examine these two patterns and their relationship to L2 proficiency. More importantly, in studies that have investigated commonality and diversity, the potential effect of responses position was not taken into consideration. Against the background of cognitive network science, the construction of large WA norms and networks has been on the rise, and some researchers have pointed out the important effect of response position, offering evidence that the lexical-semantic patterns differ from first responses to second and third responses and that as later responses are incorporated, the entire response distribution changes, with a greater variety of responses uncovered.

Based on the unsettled issues discussed above, the present study sets out to explore the effects of L2 proficiency and response position on commonality and diversity of L2 associative responses. Hopefully, the findings can provide insights into the general development of L2 lexical organization as well as implications for the selection of participants, the number of responses in creating large semantic networks and the interpretation of network characteristics as with different response positions.

LITERATURE REVIEW

Word association (WA) offers as an easy window into the organization of words in the mind and has been widely used by second language (L2) researchers to investigate the mental lexicon of L2 learners as well as development of L2 lexical knowledge. Over the past four decades, two important approaches have dominated L2 WA research: one concerned with the comparison of L2 learners' WA responses to the norms of native speakers, and the other centering around the lexical-semantic patterns derived from L2 learners' responses. Numerous studies in the literature show that as learners progress to a higher proficiency level, they tend to establish word associations that are more native-like (although the reliability of native-like stereotypy as a function of L2 proficiency has been called into question by several scholars). Also, they tend to produce more semantically related and fewer phonologically or orthographically related associations. In contrast, little attention has been paid to commonality and diversity in L2 learners' WA responses, which appear to be more objective indexes in understanding (dis)similarities in the organization of learners' lexical knowledge and have important implications in constructing large word association networks.

While early WA research differed greatly in the actual operationalization of commonality, it is generally defined as the degree a group of subjects generate common associative responses to a set of prompt words. Previous work demonstrates that L2 proficiency is positively correlated with associative commonality, indicating that more proficient learners seem to have more shared concepts in their L2 lexicon. Closely related is the notion of diversity, which is defined as the number of different associative responses by a group of subjects. While it may seem intuitive that an increase in associative commonality is accompanied by a decline in diversity, some research suggests that this is not the case, because higher proficiency learners have larger vocabularies which enable them to respond with a greater variety of associations.

Word associations (WAs) are insightful in revealing how lexical and conceptual knowledge are organized in the mental lexicon (De Deyne et al., 2017; Nelson, 1977; Stacy et al., 2006) and thus have attracted wide attention among second language (L2) researchers over the past few decades (Fitzpatrick & Thwaites, 2020; Henriksen, 2008; Jiang & Zhang, 2021; Meara, 1978). One important line in L2 research has been concerned with the degree of nativelikeness of L2 learners' WAs as compared to norms lists generated from speakers of first language (L1) (Kruse et al., 1987; Schmitt, 1998; Zareva & Wolter, 2012). The basic assumption underlying this methodology is that the associative organization of L1 speakers is highly shared and can be used as a reliable benchmark for comparison. Such notion of a shared lexicon is substantiated by the high degree of commonality in their WA responses which, as revealed by some researchers (Zareva, 2010; Zareva et al., 2005), also exists in L2 learners WAs. These studies show that learners from different proficiency levels vary significantly in how they produce common WA responses – as they become more proficient, they tend to associate in a more consistent way.

To date, however, investigations into the degree of commonality in L2 WAs have been restrained to first responses – in cases where a multiple word association task was used, first, second and third responses were often treated uniformly – whereas research on Dutch speaking adults and children demonstrates that the serial position of response has an impact on associative patterns both quantitatively and qualitatively (De Deyne & Storms, 2008b; Spatgens & Schoonen, 2020). It remains open to us whether such effect exists in L2 learners' WA as well. This study sets out to fill this research gap by investigating the effects of L2 proficiency and response position on the degree of commonality in WAs elicited from a group of Chinese undergraduate students who learn English as a foreign language (EFL).

Theoretically, the findings of this study will contribute to the development of a finer account of the mechanism underlying WA performance, since the most recent theoretical model is only concerned with the first WA responses (Eguchi et al., 2022). Methodologically, this study will inform researchers in making decisions about the number of responses they will elicit in future research, and more importantly, promote large-scale studies on L2 learners' WA responses. Pedagogically, the application of multiple WAT enables language teachers to discern more subtle differences in students' responses, which is unlikely to be revealed in single word association tasks,

and this will lead to a better understanding of the unique structure of lexical knowledge in their mental lexicon and make it easier to provide individual support in their vocabulary learning.

In most L2 studies, differences in learners' WAs were examined in light of their first responses (Fitzpatrick & Izura, 2011; Håkansson & Norrby, 2010; Namei, 2004); in cases where a multiple word association task was used, the first, second and third responses were often treated uniformly. However, previous research on Dutch speaking adults and children demonstrates that response position has a significant effect on the extent to which participants produce common (or nonunique) WAs (De Deyne & Storms, 2008a; Spatgens & Schoonen, 2020). It remains open to us whether the generation position of responses affect the degree of commonality in L2 learners' WA as well. This study sets out to fill this research gap by investigating the effects of language proficiency and response position on the degree of response commonality in WAs made by a group of Chinese undergraduate students who learn English as a foreign language (EFL). Theoretically, the findings of this study will contribute to the development of a finer account of the mechanism underlying WA performance, since the most recent theoretical model is only concerned with the first WA responses (Eguchi et al., 2022). Methodologically, this study will inform researchers in making decisions about the number of responses they will elicit in future research, and more importantly, promote large-scale studies on L2 learners' WA responses. Pedagogically, the application of multiple WAT enables language teachers to discern more subtle differences in students' responses, which is unlikely to be revealed in single word association tasks, and this will lead to a better understanding of the unique structure of lexical knowledge in their mental lexicon and make it easier to provide individual support in their vocabulary learning.

In the extant literature, an extensive amount of research has been devoted to investigating the qualitative features of WA responses. However, the quantitative features, especially the degree of response commonality, is of equal importance in xxx Early research reveals that the WA responses of L1 speakers are highly common, indicating a shared lexical network. Such characteristic, often referred to as response commonality (RC), has also been observed in L2 learners' WA responses (though to a lesser degree) and has been found to correlate reliably with language proficiency.

METHOD

Participants

Two groups of Chinese learners of English participated in the present study. At the time of data collection, they were undergraduate students of English major at a Chinese university – 30 from the first year and 30 from the third year. The majority of participants were female students, and the mean age was xx. The two groups differ in their mean age and grade level, which are taken to indicate the difference between their L2 proficiency (Agustin-Llach, 2022). In other words, the third-year students are assumed to be more proficient than their first-year counterparts in the L2.

Instruments

An online word association questionnaire was constructed on Wenjuanxing (http://www.wjx.cn) as it could be easily accessed and completed by the Chinese participants on their electronic devices. Screenshots of the questionnaire are presented in xx, xx and xx. Briefly, the welcoming page and instructions are displayed successively on two pages, followed by a practice session, in which three words – xx, xx and xx – are presented respectively. Following this are the thirty test items, which appear on separate pages. It should be noted that when participants finish 10 items, an additional page will pop up to encourage them to keep going and more importantly, to emphasize the instructions again. Such design was aimed to ensure the validity of response data and it turned out that most participants completed the online test properly as expected.

Data collection

Data collection was carried out at the beginning of January 2023. It was winter holiday, so an online test was administered. Although previous research has proved the validity of web-based word association task (De Deyne et al., 2019), it is still necessary to pilot the questionnaire specifically designed for the purpose of this study. Five English-major and 2 non-English major university students were invited to take the online WAT and provide feedback about their experience. On average, it took them 15 minutes to complete the test and none of them indicated lack of patience in taking the test. However, they suggested that: 1) it would be likely to forget the instruction and ponder a stimulus word for too long, 2) as they typed in a response, other words would automatically appear in the typing box as options, which might interfere with the responding process, and 3) the instruction did not give clear requirements on the acceptable types of responses. Based on these suggestions, several adjustments were made to the online test. First, it was made clear in the instruction that "there is no restriction on response type and any response is acceptable" and that "if you are taking the test on the mobile, please turn off the 'smart typing' before the test

to avoid interference by the input method editor. Second, when completing the tenth and the twentieth test item, a reminder would appear on a separate page, stressing that the responses should be given without thinking too much. Both the QR code and hyperlink of the questionnaire were sent to the 65 participants, with the note that they would receive a gift after they came back to school for the new semester. They were given an entire week to complete the test In the middle of January 2023, fifty-five questionnaires were received, among which 30 came from first-year students and 25 from third-year students. In total, 5 questionnaires were excluded from the dataset due to the following reasons. Three questionnaires – 1 from the first-year group and 2 from the third year – were excluded because the participants misunderstood the requirements and responded in their L1 (Chinese). One participant reported experience of study abroad and was therefore excluded as well. In addition, one participant failed to supply responses to all the stimulus words and was excluded as well. Finally, 50 questionnaires – 29 from the first-year students and 21 from the third-year students – entered the subsequent data analysis procedure. **Data analysis**

Lemmatization procedure

Before response classification and analysis of the quantitative measures, all responses should be lemmatized first. This procedure was done in compliance with the criteria proposed by Zhang (2010) and Zareva & Wolter (2012):

1. All capitalized responses were converted to lower case.
2. Inflected forms of the stimulus words were treated as the stimulus words:
    a. the plural form *-s* or *-es* of nouns (e.g., RIVER-rivers);
    b. the third-person singular form *-s* of verbs (e.g., CLIMB-climbs);
    c. the past tense, past participle, and present participle of verbs (e.g., CLIMB-climbed, THROW-thrown, VALUE-valuing);
    d. comparative and superlative forms of adjectives (e.g., OLD-older, OLD-oldest).
3. For multi-word responses, if they contained the stimulus word, only the added parts were retained. For example, in the pair “ATTEMPT-make an attempt”, only “make” was kept. Otherwise, the entire response would be treated as separate (e.g., DRY-Middle East).
4. For misspelled responses, if the spelling did not cause problem for pronunciation, it would be corrected. For example, in the pair “MENTAL-phisical”, “phisical” was

corrected as “physical”. However, in “SUPPOSE-persume”, “persume” was kept as a separate response as it sounded different from the intended response “presume”.

5. All derivational forms of stimulus words were retained as separate responses (e.g., ATTRACT-attractive, DRY-drily, etc.).

In addition, it should be noted that in Zareva & Wolter (2012, p 50), the -ed and -en forms of stimulus words (e.g., INJURE-injured, BEAT-beaten) were treated as derivational and thus kept as separate items. In this study, past tense and past participle were treated as grammatical inflection of verbs, except when the item was listed as a separate entry in *Longman Dictionary of Contemporary English* and *Collins English Dictionary*. For example, “thrown” was treated as the inflected form of “throw”, whereas “preferred” was retained as a separate response to “prefer” because “preferred” was listed as a separate adjective in dictionaries.

Scoring procedure

Since the scoring procedure was only concerned with the frequency of occurrence of responses rather than human judgement, the oft-cited subjectivity issue in studies investigating lexico-syntactic patterns could be largely avoided (Fitzpatrick, 2006; Wolter, 2002). However, special care was taken in ensuring the consistency in response coding; that is, if a response word occurred multiple times and/or in different response positions, it would be treated in the same way.

There are different ways of operationalizing commonality in L2 WA research, among which two methods have been widely used. One is to quantify commonality based on the number of responses elicited from more than one participant, and the other is concerned with the weighted measure of commonality, scoring each response based on its raw frequency of occurrence in the dataset. In this study we opted for the weighted measure, since it enabled us to reveal more subtle differences in learners’ word association behavior. To illustrate, if there are two responses, one produced by 2 participants and the other by 10 participants, both would be coded as 1 (in contrast to a score of 0 if it is an idiosyncratic response) in the first measure. However, in a weighted measure, the first response would be coded as 2 and the second as 10, and in this way, we are able to explore the extent to which each participant’s word associations conform to the group norms.

It should be noted that the two groups of subjects in our study differ in number, and it will lead to bias in estimation if we only calculate the weighted commonality measure without considering the inequality in sample size. Therefore, each score was divided by the number of subjects in the

corresponding group to obtain a new score for between-group comparisons. In addition, Levene's homogeneity test showed equal variance between the two datasets, meaning that our data satisfied the requirements for two-way analysis of variance (ANOVA).

RESULTS

Descriptive statistics of response commonality in the three positions were presented in Table 1. In general, it seems that the first responses are highly common whereas the second and third responses appear more diverse or idiosyncratic. Then, statistical analyses were conducted to determine whether the mean differences reached significance level.

Table 1. Descriptive statistics of commonality of the 1st, 2nd, and 3rd responses

	N	Minimum	Maximum	Mean	Std. Deviation
First Response	29	16.00	28.00	23.0690	2.91463
Second Response	29	9.00	22.00	16.9655	3.32182
Third Response	29	10.00	26.00	15.4828	3.87870

Statistical analyses, including Friedman tests and post-hoc Wilcoxon Signed-Rank tests, were conducted in SPSS 26. Effect size and statistical power were calculated in G*Power 3.1.

Normality test results showed that the first-response data were not normally distributed: the two-sided Kolmogorov-Smirnov statistic (the maximum absolute difference) $D(29) = 0.164$, Lilliefors $p = .044$; the Shapiro-Wilk test statistic $W = 0.922$, $p = .034$. Therefore, a non-parametric Friedman test was conducted on the three datasets instead. The analysis yielded statistically significant differences among the three datasets, X^2 (2, 29) = 40.84, $p < .001$. Effect size of Friedman test was measured by Kendall's coefficient of concordance, and results indicated a large effect, $W = .704$, $p < .001$. To further explore the differences of commonality in different response positions, pairwise comparisons were carried out through Wilcoxon Sign-Rank tests, and results are displayed in Table 2.

Table 2. Post-hoc comparison results

	First - Second	First - Third	Second - Third
Z	-4.573	-4.630	-1.908
Asymp. Sig. (2-tailed)	.000	.000	.056

The post-hoc tests revealed that the degree of commonality of responses in the first position was significantly different from both the second ($p < .001$) and the third position ($p < .001$). However, no significant difference was detected between the second responses and third responses ($p = .567$). Effect size appeared large for the comparison between the first and second responses ($r = .6$) and between the first and third responses ($r = .608$) but was comparatively small for the comparison between the second and third responses ($r = .25$). Post-hoc power analyses revealed that the power for the comparison between the first and second and between the first and third were .86 and .87 respectively, indicating sufficient statistical power.

References

Zareva, A., & Wolter, B. (2012). The 'promise' of three methods of word association analysis to L2 lexical research. *Second language research*, *28*(1), 41-67. https://doi.org/10.1177/0267658311423452

Zhang, P. (2010). A comparative study of word association patterns in Chinese EFL learners' mental lexicon. *Foreign Language Teaching and Research*, *42*(01), 9-16+80.

Agustin-Llach, M. (2022). How age and L2 proficiency affect the L2 lexicon. *System*, *104*, 102697.

De Deyne, S., Kenett, Y. N., Anaki, D., Faust, M., & Navarro, D. (2017). Large-scale network representations of semantics in the mental lexicon. In M. N. Jones (Ed.), *Big Data in Cognitive Science* (pp. 174-202). New York: Psychology Press.

De Deyne, S., Navarro, D. J., Perfors, A., Brysbaert, M., & Storms, G. (2019). The "Small World of Words" English word association norms for over 12,000 cue words. *Behavior Research Methods*, *51*(3), 987-1006. https://doi.org/10.3758/s13428-018-1115-7

De Deyne, S., & Storms, G. (2008a). Word associations: Network and semantic properties. *Behavior Research Methods*, *40*(1), 213-231. https://doi.org/10.3758/BRM.40.1.213

De Deyne, S., & Storms, G. (2008b). Word associations: Norms for 1,424 Dutch words in a continuous task. *Behavior Research Methods*, *40*(1), 198-205. https://doi.org/10.3758/brm.40.1.198

Eguchi, M., Suzuki, S., & Suzuki, Y. (2022). Lexical competence underlying second language word association tasks: Examining the construct validity of response type and response time measures. *Studies in Second Language Acquisition*, *44*(1), 112-142.

Fitzpatrick, T. (2006). Habits and rabbits, Word associations and the L2 lexicon. *EUROSLA Yearbook*, *6*, 121-145.

Fitzpatrick, T., & Izura, C. (2011). Word association in L1 and L2: An exploratory study of response types, response times, and interlingual mediation. *Studies in Second Language Acquisition*, *33*(3), 373-398. https://doi.org/10.1017/S0272263111000027

Fitzpatrick, T., & Thwaites, P. (2020). Word association research and the L2 lexicon. *Language Teaching*, *53*(3), 237-274. https://doi.org/10.1017/s0261444820000105

Håkansson, G., & Norrby, C. (2010). Environmental influence on language acquisition: Comparing second and foreign language acquisition of Swedish [Article]. *Language Learning*, *60*(3), 628-650. https://doi.org/10.1111/j.1467-9922.2010.00569.x

Henriksen, B. (2008). Declarative lexical knowledge. In D. Albrechtsen, K. Haastrup, & B. Henriksen (Eds.), *Vocabulary and Writing in a First and Second Language: Processes and Development* (pp. 22-66). Palgrave Macmillan. https://doi.org/10.1057/9780230593404

Jiang, N., & Zhang, J. (2021). Form prominence in the L2 lexicon: Further evidence from word association. *Second language research*, *37*(1), 69-90. https://doi.org/10.1177/0267658319827053

Kruse, H., Pankhurst, J., & Smith, M. S. (1987). A multiple word association probe in second language acquisition research. *Studies in Second Language Acquisition*, *9*(2), 141-154. https://doi.org/10.1017/S0272263100000449

Meara, P. (1978). Learners' Associations in French. *Interlanguage Studies Bulletin*, *3*(2), 192-211.

Namei, S. (2004). Bilingual lexical development: A persian–swedish word association study. *International Journal of Applied Linguistics (United Kingdom)*, *14*(3), 363-388. https://doi.org/10.1111/j.1473-4192.2004.00070.x

Nelson, K. (1977). The syntagmatic-paradigmatic shift revisited: A review of research and theory [Article]. *Psychological Bulletin*, *84*(1), 93-116. https://doi.org/10.1037/0033-2909.84.1.93

Schmitt, N. (1998). Quantifying word association responses: What is native-like? *System*, *26*(3), 389-401. https://doi.org/10.1016/S0346-251X(98)00019-0

Spatgens, T., & Schoonen, R. (2020). The structure of developing semantic networks: Evidence from single and multiple nominal word associations in young monolingual and bilingual readers. *Applied Psycholinguistics*, *41*(5), 1141-1169. https://doi.org/10.1017/s0142716420000430

Stacy, A. W., Ames, S. L., & Grenard, J. L. (2006). Word Association Tests of Associative Memory and Implicit Processes: Theoretical and Assessment Issues. In *Handbook of implicit cognition and addiction.* (pp. 75-90). Sage Publications, Inc. https://doi.org/10.4135/9781412976237.n6

Wolter, B. (2002). Assessing proficiency through word associations: Is there still hope? *System*, *30*(3), 315-329. https://doi.org/10.1016/S0346-251X(02)00017-9

Zareva, A. (2010). Multicompetence and L2 users' associative links: Being unlike nativelike. *International Journal of Applied Linguistics*, *20*(1), 2-22. https://doi.org/10.1111/j.1473-4192.2009.00226.x

Zareva, A., Schwanenflugel, P., & Nikolova, Y. (2005). Relationship between lexical competence and language proficiency: Variable sensitivity. *Studies in Second Language Acquisition*, *27*(4), 567-595. https://doi.org/10.1017/S0272263105050254

Zareva, A., & Wolter, B. (2012). The 'promise' of three methods of word association analysis to L2 lexical research. *Second language research*, *28*(1), 41-67. https://doi.org/10.1177/0267658311423452